

Thermodynamic basis — parameters and models, part by part

Green ammonia, 500 t/day, Sines. Companion document: *Project brief* (projectBrief.pdf).

What this document is

Every number and every model name below was read off the case's own files or off a Choupo run of it. Nothing here is a summary written from memory: each section names the file the fact lives in, so you can open it and disagree.

That matters because this case runs **three different thermodynamic worlds at once**, and a stream that crosses between them is re-priced on the other side. If you do not know which world a unit computes in, you cannot read its energy balance.

1 Three property packages, one per world

A Choupo case declares its thermophysical system in `constant/thermoPhysPropDict`, and in a sectorised case that file exists at every level: a sector's own declaration *replaces* the plant's for the units inside it.

Where	Solvent	Solutes (Henry)	Why
<code>constant/</code> (plant)	NH ₃ , pure-liquid rung	N ₂ , H ₂ , Ar	The loop and the product: the condensed phase is liquid ammonia.
<code>PURIFICATION/constant/</code>	water, pure-liquid rung	N ₂ , H ₂ , Ar, O ₂	The only liquid here is the condensate the deoxo reaction makes.
<code>RECOVERY/constant/</code>	water, pure-liquid rung	NH ₃ , N ₂ , H ₂ , Ar, O ₂	The scrubber dissolves ammonia into water.

All three declare `formulation diluteSolution`, all three carry the same six components (N2 H2 NH3 Ar water O2), and all three price the vapour with **SRK** (vapour { `fugacityModel SRK`; `}`). The liquid is where they differ.

1.1 What `diluteSolution` means here

Two reference rungs in one liquid phase, chosen per component and declared:

- the **solvent** sits on the `pureLiquid` rung — the Raoult reference, activity $\rightarrow 1$ as $x \rightarrow 1$;
- every **solute** sits on the `infiniteDilution` rung — the Henry reference, with `solutionModel henryDilute`.

This is a *declared* choice with a stated domain of validity, not a default. It holds while the solutes stay dilute. The **RECOVERY** package says so in its own header: when the scrubber liquid reaches roughly 15% ammonia by mass the speciation starts to matter, Henry

stops holding, and the package has to change. **The records for that are already in the case** — `NH3.dat` declares `aqueousSpeciation ammonia`; as an identity fact, which pulls `chemistry/NH3aq-formation.dat`, `chemistry/water-dissociation.dat` and the species NH_4^+ , OH^- , H^+ into the case — so switching formulation is a declaration change, not a data hunt.

1.2 Model boundaries, and what they do to enthalpy

Two streams cross between worlds: **Makeup** (`PURIFICATION` → `LOOP`) and **Purge** (`LOOP` → `RECOVERY`).

At a model boundary Choupo holds (T, P, z) and lets each side compute its own enthalpy. **H is the conserved truth; T is the model-dependent readout.** The step you see in the first-law ledger at those two streams is two models disagreeing about $H(T, P, z)$ — it is information, not a solver failure, and Choupo prints it rather than hiding it in a silent temperature nudge.

2 The vapour phase: SRK

Soave–Redlich–Kwong, one equation of state for the vapour in all three packages, with $k_{ij} = 0$ (announced by the run). It needs three constants per component, and this case’s records declare them as follows.

Component	MW	T_c [K]	P_c [bar]	ω	Record
NH_3	17.030	405.40	113.59	0.252	<code>constant/components/NH3.dat</code>
N_2	28.013	126.20	33.98	0.039	<code>constant/components/N2.dat</code>
H_2	2.016	33.18	13.13	−0.220	<code>constant/components/H2.dat</code>
Ar	39.948	150.86	48.98	−0.002	<code>constant/components/Ar.dat</code>
water	18.015	647.14	220.64	0.3449	<code>constant/components/water.dat</code>
O_2	31.999	154.58	50.43	0.022	<code>constant/components/O2.dat</code>

Every component also declares an **Antoine** vapour-pressure correlation and a **polynomial** ideal-gas heat capacity, each with its own validity window. Choupo announces every extrapolation outside a declared window at the end of the run — the base case reports **32** of them — so the caveat block is where you find out whether a correlation was used where its author intended.

Hydrogen at $\omega = -0.220$ and a critical temperature of 33 K is being priced by a cubic equation of state at 200 bar and 840 K. That is a long way outside where SRK was fitted, and it is a legitimate thing to say in your report. Choupo does not stop you; it tells you.

3 The liquid phase: the Henry pairs

The dilute rung needs one parameter set per (solute, solvent) pair. Choupo stores them as records under `parameters/Henry/`, each declaring its convention (**Sander-Hxp-v1**), its model (**vantHoff**), its reference fugacity, its enthalpy of dissolution, its partial molar volume where one was fitted, and **its temperature window**.

Pair	Model	H_{ref} [Pa] (298.15 K)	ΔH_{diss} [J/mol]	T range [K]	Origin
N ₂ –NH ₃	van 't Hoff	8.50×10^8	+11 040	238–305	standards
H ₂ –NH ₃	van 't Hoff	1.18×10^9	+13 600	288–333	standards
Ar–NH ₃	van 't Hoff	4.87×10^8	+9 495	243–323	fitted for this project
NH ₃ –water	van 't Hoff	9.9×10^4	–33 700	273–373	standards
N ₂ –water	van 't Hoff	8.65×10^9	–11 868	273–373	standards
H ₂ –water	van 't Hoff	7.11×10^9	–4 659	273–373	standards
Ar–water	van 't Hoff	3.96×10^9	–13 473	273–373	standards
O ₂ –water	van 't Hoff	—	—	273–373	standards

The three ammonia-solvent pairs also carry a Krichevsky–Kasarnovsky Poynting volume, $v_\infty \approx 2.3\text{--}2.4 \times 10^{-5} \text{ m}^3/\text{mol}$, and a Margules $A = 0$ (the dilute-ideal activity holds across the data). **The sign of ΔH_{diss} flips between the two solvents:** the permanent gases are *more* soluble in hot ammonia and *less* soluble in hot water. That is not a typing error, and a student who notices it has understood something.

3.1 The one pair this project had to fit: Ar–NH₃

The catalogue had no argon-in-ammonia pair, and **the absence was not neutral**. With no Henry pair, argon falls onto the Raoult rung, and its “vapour pressure” at the 250 K separator then comes from an Antoine extrapolation *above its own critical temperature* ($T_c = 150.86 \text{ K}$). That gave $K_{\text{Ar}} = 1.7$ against $K_{\text{N}_2} = 124$ — argon roughly seventy times more soluble in liquid ammonia than nitrogen — so 88 % of the argon left dissolved in the product instead of accumulating in the loop, and the purge, which is the heart of this case, was left with no physical meaning. Exit code 0 throughout.

This is the most useful single lesson in the case. A missing parameter did not produce an error. It produced a plausible number, from a correlation used far outside the window its author declared, and it quietly removed the phenomenon the whole flowsheet exists to manage.

The pair was therefore fitted for this project, and the whole derivation is reproducible: `curation/Ar-NH3/fit-Ar-NH3.py`.

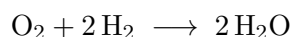
- **Primary data**, all three sets classified TENTATIVE by C. L. Young’s critical evaluation in IUPAC–NIST Solubility Data Series Vol. 4 (*Argon*), H. L. Clever (ed.), 1980, pp. 314–318: Kaminishi, G., *Kogyo Kagaku Zasshi* **68** (1965) 419 [*Int. Chem. Eng.* **5** (1965) 749, English translation] — x_{Ar} at 273.15 / 298.15 / 323.15 K, 50.8 to 199.9 bar, the anchor set because it covers exactly the loop pressure; Matouš, Sobr & Novák, *Coll. Czech. Chem. Comm.* **35** (1970) 3757 (243–303 K — the only set reaching below 273 K, i.e. covering the 250 K separator); and Michels, Dumoulin & Van Dijk, *Physica* **27** (1961) 886.
- **A correction was applied, and it is declared rather than silent.** As printed in the compilation, the Matouš values are ten times Kaminishi’s at the single overlapping point, which contradicts the critical evaluation’s own statement that the studies agree. Read as $10^1 x_{\text{Ar}}$ they fall within $\pm 4 \%$ of the van 't Hoff line at every temperature. That reading is adopted and marked *to be confirmed against the original 1970 paper*.

- **Independent validation, held out of the fit.** Michels 1961 — a different laboratory, a different method, with y_{Ar} measured — gives $H = 4.56 \times 10^8 \text{ Pa}$, -6% from the adopted value.
- **Family consistency.** $v_\infty = 23 \text{ cm}^3/\text{mol}$ against 23 (N_2) and 24 (H_2) in the same solvent; argon is 1.8 times more soluble than nitrogen in liquid ammonia, against 2.2 times in water. The same order of magnitude, not seventy times.

This pair is the case's one piece of original curation, and it is the model for what Choupo asks of any parameter you add: primary sources, a stated correction, a held-out validation set, a declared window, and a script that reproduces the fit.

4 The reaction: the deoxo bed

Declared in `PURIFICATION/constant/reactions`:



with `limitingReactant O2` and `conversion 1.0` in `PURIFICATION/Deoxo` — hydrogen is in roughly 700-fold excess, so complete conversion of the oxygen is the real specification of a deoxo bed (outlet O_2 below 1 ppm).

The heat of reaction is not declared anywhere and must not be. Choupo computes it as $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$ from each species' own formation enthalpy in its `standardThermochemistry` block. A `dH_rxn` key in a reactions dict is an announced, cross-checked *override* for toy systems that lack formation data — never a primary input. One enthalpy datum, one place.

Measured on the base case: **2.594 kmol/h** of oxygen removed, releasing **348.7 kW**, and making **81.05 kg/h** of water — which is the entire reason the chiller, the knock-out drum and the dryer exist downstream.

5 The converter: Gibbs equilibrium, not kinetics

`LOOP/Converter` is a `gibbsReactor` in mode `adiabatic` at 200 bar. It minimises Gibbs energy over the declared species (N_2 , H_2 , NH_3 , Ar) subject to the declared element balances (N, H, Ar), and finds the outlet temperature that closes the energy balance.

- **There is no rate law and no catalyst in this model.** The result is the equilibrium the bed would reach given infinite contact time. A real converter approaches it; how closely is a catalyst question this case does not ask.
- Argon is carried as an inert species: it takes part in no reaction but occupies volume and dilutes the reactants, which is exactly its role in the real loop.
- Measured: outlet at **839.0 K** carrying **11.01 mol %** ammonia.

An open item in Choupo, stated here rather than discovered by you. The adiabatic Gibbs reactor solves its energy balance on the ideal-gas enthalpy surface while its streams are priced with the SRK departure. At 200 bar those are not the same surface, and the difference lands in the first-law ledger. This is a known engine-level issue, not a defect in your case.

6 The dryer: adsorption equilibrium

PURIFICATION/Dryer is a `tsaTwinBed`: 1500 kg of 4A molecular sieve per column, 12 h of adsorption per bed, regenerated at 553 K with 0.5 % of the product gas as purge. Its equilibrium comes from `PURIFICATION/constant/parameters/adsorption/equilibria/molSieve4A`.

The adsorbent's bulk density (700 kg/m³) and solid heat capacity (920 J/kg K) are declared in `PURIFICATION/constant/adsorbents/` and **marked, in the record itself, as a case-local teaching anchor rather than curated data**: representative material-class values for a commercial 4A sieve in 3 mm beads, not design data, and explicitly not to be promoted to the standards catalogue. Measured: the bed runs at **52.7 %** of its working capacity, which is a design margin you may argue with.

7 Sizing and costing parameters

Declared in `system/postDict`. **Every one of these is a design decision, not a physical constant** — they are the numbers your report has to defend.

Item	Model	Declared	Material
PURIFICATION.N2Boost	compressor	P_{des} 35 bar	carbonSteel
PURIFICATION.SynComp	compressor	P_{des} 250 bar	SS316
LOOP.RecycleComp	compressor	P_{des} 250 bar	SS316
PURIFICATION.N2Cooler	shellTubeHX	U 500, LMTD 60	carbonSteel
LOOP.FEHE	shellTubeHX	U 550, LMTD 150	SS316
LOOP.WaterCooler	shellTubeHX	U 700, LMTD 60	carbonSteel
LOOP.Converter	vessel	V 28 m ³ , L/D 6	SS316
LOOP.Separator	vessel	τ 45 s, L/D 3	SS316
PRODUCT.Letdown	vessel	τ 60 s, L/D 3	carbonSteel

U in W/(m²·K), LMTD in K. Exchanger area follows $A = Q/(U \text{ LMTD})$ with Q taken from the unit's own solved duty; vessel volume follows $V = Q_{\text{gas}}\tau$ except for the Converter, whose volume is declared outright. **Choupo states the basis of every size it computes**, on the console and in each specification sheet, so a declared size never looks like a computed one.

Costing: Turton correlations, CEPCI 820 (2026) against 397 (2001), USD/EUR 0.92, contingency and fee $1.18 \times C_{\text{BM}}$, site factor 0.50. Economics: discounted cash flow, 20-year life, 8 % discount rate, 21 % tax, straight-line depreciation over 10 years, working capital 15 % of FCI, stream factor 0.913.

7.1 What the run says about its own sizing

- **The gas volumetric flow in the vessel sizer is NRT/P** , ideal gas — printed in the sizer's own basis line, next to the number. At 200 bar in the loop this is the one place in the chain where the stated basis and the case's thermodynamic package disagree.
- **SS316 is rated 100 bar** in `constant/assets/SS316.dat` and the Converter and Separator are designed at 240 bar. Warned per vessel, priced anyway.
- **PRODUCT.Letdown at $V_R = 0.125 \text{ m}^3$** falls below the Turton correlation's range [0.3, 520]; its cost is extrapolated and says so.

8 Where every fact in this document lives

Fact	File
Package, per world	[<SECTOR>/]constant/thermoPhysPropDict
Critical constants, Antoine, c_p	constant/components/*.dat
Henry pairs (case-local)	constant/parameters/Henry/*.dat
Ar–NH ₃ fit, reproducible	curation/Ar-NH3/
Aqueous chemistry (unused at base)	constant/chemistry/, constant/species/
Reaction	PURIFICATION/constant/reactions
Adsorbent	PURIFICATION/constant/adsorbents/
Topology and units	[<path>/]system/flowsheetDict
Initial state, per stream	0/<LEVEL>/<stream>
Sizing, costing, economics	system/postDict, constant/economics
Materials	constant/assets/*.dat

Run the case and read **converged/** for the solved state, **design/** for one specification sheet per physical item, and **reports/balances/** for the mass, element and energy ledgers. The end-of-run caveat block lists every extrapolation, every unverified record and every assumed default the run actually used.